EVERY ALGEBRA IS A DIRECTED COLIMIT OF FINITELY PRESENTED ALGEBRAS, WITH SURJECTIVE TRANSITIONS IN THE FINITE TYPE CASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The second proof inside the counted span is the explicit construction; the first proof is a citation of a categorical statement, and both stay verbatim inside the span.

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Allowed standard prerequisites (editorial).

- Finite presentation as finitely many generators and relations. Used in the explicit construction in the second proof (lines 32067-32094). Supplies the finitely presented approximation attached to a finite piece of data.
- Finite type as a quotient of a polynomial algebra. Used when arranging the transition maps to be surjective (lines 32080-32093). Lets the index set be thinned so that the transition maps become surjections.

Source reference. Original lemma 0032: sources/stacks-chow/tags.txt, source lines 119–119. Navigation pointer for \ref{categories-lemma-directed-category-system} used inside the counted span (line 32065); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HA: sources/stacks-algebra/pinned-original-source.tex, source lines 1847–1847. Navigation pointer for \ref{lemma-module-colimit-fp} used inside the counted span (line 32069); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BUF: sources/stacks-algebra/pinned-original-source.tex, source lines 31988–31988. Navigation pointer for \ref{lemma-ring-colimit-fp-category} used inside the counted span (line 32064); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to A$ be a ring map. There exists a directed system A_λ of R-algebras of finite presentation such that $A = \operatorname{colim}_\lambda A_\lambda$. If A is of finite type over R we may arrange it so that all the transition maps in the system of A_λ are surjective.*

Proof. The first proof is that this follows from Lemma 0BUF and Categories, Lemma 0032.

Second proof. Compare with the proof of Lemma 00HA. Consider any finite subset $S \subset A$, and any finite collection of polynomial relations E among the elements of S. So each $s \in S$ corresponds to $x_s \in A$ and each $e \in E$ consists of a polynomial $f_e \in R[X_s; s \in S]$ such that $f_e(x_s) = 0$. Let $A_{S,E} = R[X_s; s \in S]/(f_e; e \in E)$ which is a finitely presented R-algebra. There are canonical maps $A_{S,E} \to A$. If $S \subset S'$ and if the elements of E correspond, via the map $R[X_s; s \in S] \to R[X_s; s \in S']$, to a subset of E', then there is an obvious map $A_{S,E} \to A_{S',E'}$ commuting with the maps to A. Thus, setting Λ equal the set of pairs (S, E) with ordering by inclusion as above, we get a directed partially ordered set. It is clear that the colimit of this directed system is A.

For the last statement, suppose $A = R[x_1, \ldots, x_n]/I$. In this case, consider the subset $\Lambda' \subset \Lambda$ consisting of those systems (S, E) above with $S = \{x_1, \ldots, x_n\}$. It is easy to see that still $A = \operatorname{colim}_{\lambda' \in \Lambda'} A_{\lambda'}$. Moreover, the transition maps are clearly surjective. $\square$